
27.5 LET US SUM UP

In this unit, you learnt the concepts of joint distribution and the associated marginal distribution. You have also been introduced to the three most elementary and most used distributions in the theory of probability namely, Binomial, Poisson and Normal. First two of these are discrete and the last one is continuous. A distribution is defined mostly by its moments; therefore, having understood the concepts concerning moments we discussed how these moments are used to characterize a distribution.

27.6 KEY WORDS

- Binomial Distribution** : A discrete probability distribution satisfying the following conditions is a binomial distribution:
- 1) It involves finite repetition of identical trials.
 - 2) Each trial has two possible outcomes: success and failure.
 - 3) Trials are independent of each other.
 - 4) The probability of the outcomes (success and failure) does not change from one trial to another.
- Normal Distribution:** : It is a continuous probability distribution with the following properties:
- 1) It is symmetrical about its mean;
 - 2) All the measures of central tendency for this distribution is the same, i.e., for normal distribution Mean = Mode = Median; and
 - 3) A variable following normal distribution can take any value within the range $(-\infty, \infty)$
- Poisson Distribution** : A discrete probability distribution which is the limiting form of the binomial distribution, provided
- 1) The number of trials is very large is fact tending to infinity.
 - 2) The probability of success in each trial is very small; tending to zero.
- Probability Distribution** : Probability distribution of a random variable is a statement specifying the set of its possible values together with the respective probabilities.

27.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

$$1) \quad f_X(x) = \int_0^{3-3x} \frac{x+y}{4} dy = \frac{9}{8} - (3/2)x + \frac{3}{8}x^2, 0 < x < 1$$

$$f_Y(y) = \int_0^{1-(y/3)} \frac{x+y}{4} dx = \frac{1}{8} + \frac{1}{6}y - \frac{5}{72}y^2, 0 < y < 3$$

$$2) \text{ Hint: } g(x) = g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_2^{\frac{1}{3}} \frac{2}{3}(x+2y) dy = \frac{2}{3}(x+1) \text{ for } 0 < x < 1 \text{ and } g(x) = 0$$

elsewhere

Likewise, you can find out $h(y)$

Check Your Progress 2

- 1) As the formula for mean and variance of binomial distribution are given by $n.p$ and $n.p.(1-p)$ respectively, we get the following two equations

$$n.p = 4 \dots \dots \dots (1)$$

$$n.p.(1-p) = 8/3 \dots \dots \dots (2)$$

Solving these two equations we get $n = 12$ and $p = 1/3$.

- 2) See Section 27.4.2.

- 3) Hints: x follows a Binomial distribution and the event of a candidate passing the examination as success. Thus the probability of success:
 $f(x) = {}^6C_x (60/100)^x (40/100)^{6-x}$. Find out $f(4) + f(5) + f(6)$

Ans: 1701/3125

- 4) Hints: Let the $f(x) = {}^{10}C_x (P)^x (Q)^{10-x}$: and it is given that $f(5) = 2f(4)$

$$\text{Ans: } \left(\frac{3}{8}\right)^{10}$$

- 5) $\lambda = 10 \times 0.1 = 1$, hence the probability of three defectives in the sample is given by

$$f(3) = e^{-1} \times 1^3 / 3! = 0.061$$

- 6) The number of defectives follows a binomial distribution. Since $p = .02$ and $n = 100$ which is very large, making up, $n.p = \lambda = 100 \times .02 = 2$, a finite quantity, we use the poisson approximation to binomial distribution.

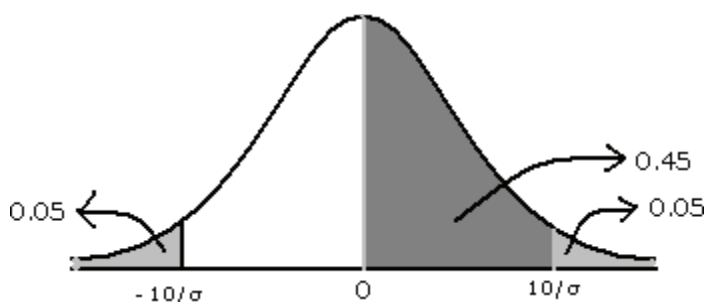
$$P(\text{of 3 or more defectives}) = 1 - [f(0) + f(1) + f(2)] = 1 - e^{-2} [1 + 2 + 2^2/2!] = 0.325$$

- 7) Substituting $x = 7$, $n = 10$, and $p = 0.8$ into the formula for binomial distribution, we get

$$b(7; 10, 0.8) = {}^{10}C_7 (0.8)^7 (0.2)^3 = 0.2 \text{ (approximately)}$$

Check Your Progress 3

- 1) Clearly, the random variable denoting the weight, say X , of the students follow a normal distribution, mean $(X) = 151$; $\text{Var}(X) = 15^2$
 - i) Proportion of students whose weight lie between 120 and 155 lbs
= Area under the standard normal curve between the vertical lines at the standardized values; $z = (120 - 151)/15 = -2.07$ and $z = (155 - 151)/15 = 0.27$
 $P(120 \leq X \leq 155) = \Phi(0.27) - \Phi(-2.07) = \Phi(0.27) - \{1 - \Phi(2.07)\}$
 $= 0.6064 - 1 + 0.9808 = 0.5872$
 - ii) $P(X > 155) = 1 - P(X \leq 155) = 1 - \Phi(0.27) = 0.3936$
- 2) The probability that X takes values greater than 60 is 5% or 0.05. This must be the area under the standard normal curve to the right of the ordinate at the standardized value $z = (60 - 50)/\sigma = 10/\sigma$



Since the area to the right of $z = 0$ is 0.05 and the area between $z = 0$ and $z = 1.64$ is given to be 0.45, the area to the right of $z = 1.64$ is $.5 - .45 = .05$. Thus, we get $10/\sigma = 1.64$, or, $\sigma = 6.1$.

27.8 EXERCISES

- Q1.** Find the binomial distribution of getting a six in three tosses of an unbiased dice.

Ans: Let X be the random variable of getting six. Then X can be 0, 1, 2, 3.

Here, $n = 3$

p = Probability of getting a six in a toss = $\frac{1}{6}$

q = Probability of not getting a six in a toss = $1 - \frac{1}{6} = \frac{5}{6}$

$$P(X = 0) = {}^nC_r p^r q^{(n-r)} = {}^3C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^{3-0}$$

$$= 1 \times 1 \times \frac{125}{216} = \frac{125}{216}$$

$$P(X = 1) = {}^nC_r p^r q^{(n-r)} = {}^3C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^{3-1}$$

$$= 3 \times \frac{1}{6} \times \frac{25}{36} = \frac{25}{72}$$

$$P(X = 2) = {}^nC_r p^r q^{(n-r)} = {}^3C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{3-2}$$

$$= 3 \times \frac{1}{36} \times \frac{5}{6} = \frac{5}{72}$$

$$P(X = 3) = {}^nC_r p^r q^{(n-r)} = {}^3C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{3-3}$$

$$= 1 \times \frac{1}{216} \times 1 = \frac{1}{216}$$

The required binomial distribution of X is:

X	0	1	2	3
P(X)	125/216	25/72	5/72	1/216

Q2. There are four fused bulbs in a lot of 10 good bulbs. If three bulbs are drawn at random with replacement, find the probability distribution of the number of fused bulbs drawn.

Ans: This is a problem of binomial distribution as the event of drawing a fused bulb is independent.

$$p = P(\text{drawing a fused bulb}) = \frac{4}{(10 + 4)} = \frac{2}{7}$$

$$q = P(\text{drawing a bulb which is not fused}) = 1 - \frac{2}{7} = \frac{5}{7}$$

X = event of drawing a fused bulb

X can take up the values 0, 1, 2, 3

$$P(X = 0) = P(\text{getting zero fused bulbs in all draws})$$

$$= {}^nC_r p^r q^{(n-r)}$$

$$= {}^3C_0 \left(\frac{2}{7}\right)^0 \left(\frac{5}{7}\right)^{(3-0)}$$

$$= 1 \times 1 \times \left(\frac{125}{343}\right) = \frac{125}{343}$$

$$P(X = 1) = P(\text{getting one time fused bulb})$$

$$= {}^nC_r p^r q^{(n-r)}$$

$$= {}^3C_1 \left(\frac{2}{7}\right)^1 \left(\frac{5}{7}\right)^{(3-1)}$$

$$= 3 \times \left(\frac{2}{7}\right) \times \left(\frac{25}{49}\right) = \frac{150}{343}$$

$$P(X = 2) = P(\text{getting two times fused bulbs})$$

$$= {}^nC_r p^r q^{(n-r)}$$

$$= {}^3C_2 \left(\frac{2}{7}\right)^2 \left(\frac{5}{7}\right)^{(3-2)}$$

$$= 3 \times \frac{4}{49} \times \left(\frac{5}{7}\right) = \frac{60}{343}$$

$$P(X = 3) = P(\text{getting three times fused bulb})$$

$$= {}^nC_r p^r q^{(n-r)}$$

$$= {}^3C_3 \left(\frac{2}{7}\right)^3 \left(\frac{5}{7}\right)^{(3-3)}$$

$$= 1 \times \frac{8}{343} \times 1 = \frac{8}{343}$$

The required probability distribution:

X	0	1	2	3
P(X)	125/343	150/343	60/343	8/343

Q3. A life insurance salesman sells on the average 33 life insurance policies per week. Use Poisson's law to calculate the probability that in a given week he will sell

- Some policies
- 2 or more policies but less than 5 policies.
- Assuming that there are 5 working days per week, what is the probability that in a given day he will sell one policy?

Ans. Here, $\mu = 3$

- "Some policies" means "1 or more policies". We can work this out by finding 1 minus the "zero policies" probability:

$$P(X > 0) = 1 - P(x_0)$$

$$\text{Now } P(X) = e^{-\mu} \mu^x / x!$$

$$\text{So, } P(x_0) = e^{-3} 3^0 / 0!$$

$$= 4.9787 \times 10^{-2}$$

Therefore, the probability of 1 or more policies is given by: Probability = $P(X \geq 0)$

$$= 1 - P(x_0)$$

$$= 1 - 4.9787 \times 10^{-2}$$

$$= 0.95021$$

- The probability of selling 2 or more, but less than 5 policies is: $P(2 \leq X < 5)$

$$= P(x_2) + P(x_3) + P(x_4)$$

$$= (e^{-3} 3^2 / 2!) + (e^{-3} 3^3 / 3!) + (e^{-3} 3^4 / 4!)$$

$$= 0.61611$$

- Average number of policies sold per day: $3/5 = 0.6$

$$\text{So, on a given day, } P(X) = e^{-0.6} (0.6)^1 / 1!$$

$$= 0.32929$$

Q4. For some computers, the time period between charges of the battery is normally distributed with a mean of 50 hours and a standard deviation of 15 hours. Rohan has one of these computers and needs to know the probability that the time period will be between 50 and 70 hours.

Ans: Let x be the random variable that represents the time period.

Given Mean, $\mu = 50$

and standard deviation, $\sigma = 15$

To find probability that x is between 50 and 70 or $P(50 < x < 70)$:

By using the transformation equation, we know;

$$z = (X - \mu) / \sigma$$

$$\text{For } x = 50, z = (50 - 50) / 15 = 0$$

$$\text{For } x = 70, z = (70 - 50) / 15 = 1.33$$

$$P(50 < x < 70) = P(0 < z < 1.33) = [\text{area to the left of } z = 1.33] - [\text{area to the left of } z = 0]$$

From the table we get the value, such as;

$$P(0 < z < 1.33) = 0.9082 - 0.5 = 0.4082$$

The probability that Rohan's computer has a time period between 50 and 70 hours is equal to 0.4082.

Q5. The speeds of cars are measured using a radar unit, on a motorway. The speeds are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr. What is the probability that a car selected at chance is moving at more than 100 km/hr?

Ans: Let the speed of cars be represented by a random variable 'x'.

Now, given mean, $\mu = 90$ and standard deviation, $\sigma = 10$.

To find probability that x is higher than 100 or $P(x > 100)$:

By using the transformation equation, we know;

$$z = (X - \mu) / \sigma$$

Hence,

$$\text{For } x = 100, z = (100 - 90) / 10 = 1$$

$$P(x > 100) = P(z > 1) = [\text{total area}] - [\text{area to the left of } z = 1]$$

$$P(z > 1) = 1 - 0.8413 = 0.1587$$